

Southeastern Section - 68th Annual Meeting - 2019

Paper No. 16-2

Presentation Time: 1:00 PM-5:00 PM

FORECASTING OF WATER DISCHARGE USING TIME SERIES DECOMPOSITION OF HYDROGEOLOGICAL AND ATMOSPHERIC VARIABLES TO IDENTIFY HIGH RISK PERIODS OF FLOODING IN SCHENECTADY, NEW YORK

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Flood events commonly occur in winter periods due to ice jams or during periods of high intensity precipitation during the spring and summer seasons. To further increase accuracy flood forecasting in Schenectady County, New York on the Mohawk River time series decomposition and a time series regression (TSR) model were used. By examining climatic and hydrogeological variables decomposition of the data was applied to separate the variables different frequencies. The decomposed data created 14,235 dummy variables that were decomposed into the long-term, seasonal-term, and short-term components using the Kolmogorov-Zurbenko filter. The prediction model had lag operators applied to increase the power of prediction compared to raw decomposed data and to minimize the collinearity between the variables. The time series regression in the long-term achieved a coefficient of determination of 93%, the seasonal term had 58% and the short-term had a coefficient of determination of 41%. All values found for time series regression had higher values than the multi-linear regression (MLR) technique. The time series regression model had an overall accuracy increased to 73%. The lag operators selected provide the strongest correlation with the water discharge time series and provide an increase in the accuracy of the model by two times. The predictor lag variables applied may hold values and dates at which the independent variables contribute to the increase in water discharge. Due to the cyclical nature of the water table depletion in the winter and water table replenishment in spring and summer months, the TSR model does perform better than the MLR model for predicting an increase in water discharge.

Session No. 16--Booth# 2

[T12. Natural Hazards Prediction \(Posters\)](#)

Thursday, 28 March 2019: 1:00 PM-5:00 PM

[Carolina Ballroom \(Francis Marion Hotel\)](#)

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